
IAL Physics Unit 5 (WPH15): A* Revision Guide

Thermodynamics, Radiation, Oscillations and Cosmology

Compiled from 15 WPH15 papers and their mark schemes (October 2020 – June 2026). Mark-scheme wording is quoted where the exact phrase earns the mark.

1. The paper and where the marks are

Format. 1 hour 45 minutes, 90 marks. Section A: 10 multiple-choice questions (1 mark each). Section B: about 10 structured questions (80 marks), usually including one asterisked 6-mark extended-response question. Topics are mixed in context questions (e.g. a satellite question tests orbits *and* thermal energy).

Topic	How often (15 papers)	What is actually asked
Gas laws and kinetic theory	Every paper	$pV = NkT$ (often with mass of molecule, or a % change); $\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$; 6-mark pressure explanation.
Heating, latent heat, power	Almost every paper	$mc\Delta\theta + mL$ combined with $P = E/t$, efficiency, or energy balance (ice in drink). Ends with “assess the claim”.
Radioactive decay	Every paper	Balanced β^-/α equations; $\lambda = \ln 2/t_{1/2}$; $A = \lambda N$; $A = A_0 e^{-\lambda t}$; power = activity $\times$ energy per decay.
Mass defect, binding energy, fission/fusion	Most papers	Δm in u $\rightarrow$ kg $\rightarrow E = c^2\Delta m \rightarrow$ MeV; B.E./nucleon graph; conditions for fusion (frequent 6-marker).
Gravitational fields and orbits	Most papers	$g = GM/r^2$; $V = -GM/r$; $\Delta E = m\Delta V$; orbital period, orbits per day, geostationary height; deriving $T^2 \propto r^3$.
Stars: L , T , I	Almost every paper	Wien, Stefan-Boltzmann, inverse square law chained together to find T , r , L or d .
H-R diagram and stellar evolution	Most papers	Label axes (reverse log T); place Sun/regions; evolution 6-marker; cluster age.
Distances and cosmology	Most papers	Parallax method and its limit; standard-candle method; Doppler z ; Hubble’s law; age = $1/H_0$; dark matter; fate of universe.
SHM, resonance, damping	Every paper	SHM definition (2 marks); $T = 2\pi\sqrt{m/k}$ or $\sqrt{l/g}$; $v_{\max} = \omega A$; E_k graphs; resonance and dampers.

2. How A* scripts collect marks

RULES TAKEN FROM THE MARK SCHEMES

1. **“Show that”**: give the final answer to **one more significant figure** than the value printed. A bare answer scores nothing. If you cannot get the value, **use the printed value** in the next part (examiners provide “use of show-that value gives ...”).
2. **Units**: a missing or wrong unit normally loses the **final** mark. Kelvin is K, never °K.
3. **$g = 9.81 \text{ N kg}^{-1}$** (9.8 accepted). Using 10 loses a mark.
4. **$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$** . One recent scheme explicitly rejected 1.67×10^{-27} . Do not round intermediate values in mass-defect work; the difference of nearly equal numbers needs every digit.
5. **“Assess / is the claim valid?”** The last mark needs three things: your calculated value, a **numerical comparison** with the value in the question (“0.35 % < 0.5 %”), and a **conclusion**. A calculation alone never gets that mark.
6. **Graph reading**: read values within the stated tolerance; use a **large triangle** for gradients; read **several periods** and divide when finding T .
7. **Specific words are underlined in the schemes**. Learn the phrase bank in section 9. For example, “equilibrium *position*” (just “equilibrium” is rejected); “*nuclei*/protons” (not atoms/particles/molecules) in fusion answers; “*measure* intensity” (not “calculate”).

6-MARK EXTENDED RESPONSE: HOW IT IS SCORED

Six indicative content (IC) points. Marks for content: 6 IC → 4, 5–4 IC → 3, 3–2 IC → 2, 1 IC → 1. Up to 2 extra marks for linkage. You can only get the full 2 linkage marks if you have at least 5 IC points, so **6/6 requires all six points in a connected chain**.

- Write **6–7 numbered sentences**, each a cause → effect step joined with “so”, “therefore”, “because”.
- Use each key noun exactly (molecules, nuclei, core, natural frequency). Do not drift into a related topic.
- Model answers for every 6-marker set so far are in section 8.

3. Thermal physics

$\Delta E = mc\Delta\theta$	$\Delta\theta$ is a temperature difference (°C or K, same number).
$\Delta E = L\Delta m$	Fusion (melting/freezing) or vaporisation (boiling/condensing). T constant during the change.
$pV = NkT$	T in kelvin ($\theta + 273$). N = number of molecules = total mass ÷ mass of one molecule.
$p_1V_1/T_1 = p_2V_2/T_2$	Fixed N . For a sphere, $V \propto r^3$ (so $T_2 = T_1(r_2/r_1)^3$ at constant p).

$\frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$	Mean KE per molecule depends only on T . Doubling $T \rightarrow c_{\text{rms}} \times \sqrt{2}$. Same T : $m_1/m_2 = \langle c_2^2 \rangle / \langle c_1^2 \rangle$.
Ideal gas internal energy $U = N \times \frac{3}{2}kT = \frac{3}{2}pV$	Ideal gas: no molecular PE, so $U =$ total random KE. U is not pV ; the scheme refuses NkT .

Definitions the schemes reward. *Internal energy*: the sum of the (randomly distributed) kinetic and potential energies of the molecules. *Specific heat capacity*: energy needed to raise 1 kg by 1 K. *Specific latent heat*: energy needed to change the state of 1 kg without a change of temperature. *Absolute zero*: the temperature at which molecules have minimum (zero) kinetic energy.

Cooling curve (wax, water). While the temperature falls, molecular KE decreases. On the flat section, T is constant so KE is constant, but molecular **PE decreases** as bonds form. Internal energy decreases throughout.

ROUTINE: ENERGY-BALANCE (MIXING) PROBLEMS

Write the balance in words first; it is the first marking point.

energy lost by hot substance = energy to melt ice (mL) + energy to warm the meltwater from 0 °C to θ ($mc(\theta - 0)$)

EXAMPLE (OCT 2020). 0.325 kg water at 85.0 °C, 0.0625 kg ice at 0 °C. $0.325 \times 4190 \times (85.0 - \theta) = 0.0625 \times 3.33 \times 10^5 + 0.0625 \times 4190 \times \theta \rightarrow 1362\theta + 262\theta = 1.16 \times 10^5 - 2.08 \times 10^4 \rightarrow \theta = 58.8$ °C. Keep θ as an unknown on both sides and collect terms; do not guess a final temperature.

EXAMPLE (JUN 2024). Airship: $N = pV/kT = (1.08 \times 10^5 \times 7020)/(1.38 \times 10^{-23} \times 298) = 1.84 \times 10^{29}$; mass = $N \times 6.64 \times 10^{-27} = 1220$ kg. A “mass of gas” question is always $pV = NkT$ followed by $M = Nm$.

Other recurring contexts.

- **Kettle/microwave efficiency:** useful $P = mc\Delta\theta/t$; electrical $P = VI$; efficiency = useful $\div$ input.
- **Lid on a pan:** less steam escapes, or steam condenses on the lid and returns its latent heat. Either way, less energy is transferred to the surroundings.
- **Coolant flow:** mass per second $\times c \times \Delta\theta =$ power. Water has a higher c , so for the same power the temperature change is smaller.
- **Specific latent heat practical errors** ($L = Pt/m$): energy lost to the surroundings makes the calculated L **too large**; the heater running longer than the recorded t makes it **too small** (Jun 2026 MCQ).
- **Thermistor calibration (6 marks, Oct 2024):** start with ice + water at 0 °C $\rightarrow$ thermometer beside the thermistor $\rightarrow$ read the resistance on an ohmmeter at each temperature $\rightarrow$ heat (water bath/immersion heater) and **stir** $\rightarrow$ readings about every 10 °C $\rightarrow$ plot R against θ .

4. Nuclear radiation

$\lambda t_{1/2} = \ln 2$	Keep λ and t in matching units (s^{-1} with s, year^{-1} with years).
$A = -dN/dt = \lambda N$	N from mass: $N = \text{mass} \div (\text{nucleon number} \times 1.66 \times 10^{-27} \text{ kg})$.
$N = N_0 e^{-\lambda t}$, $A = A_0 e^{-\lambda t}$	Solve for t : $t = \ln(A/A_0)/(-\lambda)$. For “90 % decayed” use $N/N_0 = 0.10$.
$\Delta E = c^2 \Delta m$	1 eV = 1.60×10^{-19} J, so 1 MeV = 1.60×10^{-13} J. 1 u $\approx$ 931-934 MeV/ c^2 (show-that, Jan 2022).
Power from a source	$P = A \times (\text{energy per decay in J})$.
Absorption $I = I_0 e^{-\mu x}$	$\ln I = \ln I_0 - \mu x$: gradient = $-\mu$. Half-value thickness $x_{1/2} = \ln 2/\mu$. Absorption is exponential, not linear.

Decay equations (2 marks: top line, bottom line)

β^- : ${}^A_Z X \rightarrow {}^A_{Z+1} Y + {}^0_{-1} \beta + {}^0_0 \bar{\nu}_e$. **Always include the electron antineutrino.** α : ${}^A_Z X \rightarrow {}^{A-4}_{Z-2} Y + {}^4_2 \alpha$. Check that nucleon numbers and proton numbers both balance.

ROUTINE: ENERGY RELEASED / BINDING ENERGY (4-6 MARKS)

1. Mass difference: (parent) – (daughter + α). For binding energy: $(Z \times m_p + (A-Z) \times m_n) - m_{\text{nucleus}}$. Include every nucleon.
2. Convert u $\rightarrow$ kg ($\times 1.66 \times 10^{-27}$).
3. $\Delta E = c^2 \Delta m$ (in J).
4. Convert J $\rightarrow$ MeV ($\div 1.60 \times 10^{-13}$). For B.E. per nucleon, divide by A .

Masses given in MeV/ c^2 or GeV/ c^2 : subtract directly and the answer is already in MeV. (Jun 2021: $1.74248 \text{ GeV}/236 = 7.38 \text{ MeV}$ per nucleon.)

EXAMPLE (OCT 2021). Ra-228 α decay: $\Delta m = 228.02873 - 224.02021 - 4.00260 = 5.92 \times 10^{-3} \text{ u} = 9.83 \times 10^{-30} \text{ kg} \rightarrow 8.85 \times 10^{-13} \text{ J} = 5.53 \text{ MeV}$. α KE share (momentum conserved, two bodies): $E_\alpha = (m_{\text{daughter}}/m_{\text{parent}}) \times Q = (224/228) \times 5.53 = 5.43 \text{ MeV}$.

Explanations that recur

Why the α has almost all the KE	Momentum is conserved. The daughter nucleus is much more massive, so its recoil velocity is much smaller, so its KE ($p^2/2m$) is much smaller.
Why β energies form a continuous spectrum	Energy and momentum are shared between three particles (β , antineutrino, recoil nucleus). The split between β and antineutrino is random, so the β energy varies up to a maximum.
Random vs spontaneous	<i>Random</i> : cannot predict which nucleus will decay next or when. <i>Spontaneous</i> : not affected by external conditions (temperature, pressure, chemistry). The MCQs swap these words.

Identify the radiation	Paper causes no drop in count → no α . A few mm of aluminium drops the count to background → it is β , and there is no γ (γ would pass through Al).
Cloud-chamber tracks	Track thickness depends on ionising ability: α is strongly ionising (thick), β less so (thin). Track straightness depends on mass: α is massive (straight), β has low mass (twisted, deflected).
Background count	Measure it without the source, then subtract it from every reading (otherwise there is a systematic error). Count for longer, or repeat and average, to reduce the effect of random variation. Correct for background <i>before</i> applying the inverse square law.
Safety with an α source	Use tongs: α has a range of only a few cm in air, and intensity falls with distance. Minimise exposure time because the dose is cumulative. α is dangerous mainly if inhaled/ingested : it is highly ionising and damages living cells.
Beta thickness gauge	β partly passes through paper, so the count rate changes as the thickness changes (α is all absorbed; γ is barely affected).

Binding energy per nucleon, fission and fusion

- The curve rises steeply from H, peaks at **Fe-56** (about 8.8 MeV), and falls slowly for heavy nuclei. Higher B.E. per nucleon means a more stable nucleus.
- Both fission and fusion increase B.E. per nucleon**, so total mass decreases and energy is released. Energy released $\approx$ (increase in B.E. per nucleon) $\times$ (number of nucleons). U fission releases much more per event because it has many more nucleons (Jan 2026).
- Graph method (Jun 2023): $E = \Sigma(\text{B.E. of products}) - \Sigma(\text{B.E. of reactants})$, where B.E. = (B.E. per nucleon) $\times$ A. ${}^2\text{H} + {}^3\text{H} \rightarrow {}^4\text{He}$: $4(7.0) - [2(1.1) + 3(2.8)] = 17.4 \text{ MeV}$.
- Fusion conditions**: very high temperature (nuclei have enough KE to overcome electrostatic repulsion and get close enough to fuse) and very high density (collision rate high enough to sustain fusion). On Earth the plasma would cool or melt any container it touched, so **magnetic fields** are used to contain it.

5. Gravitational fields and orbits

$F = Gm_1m_2/r^2$	r = centre-to-centre distance. For an orbit, r = radius of planet + height .
$g = F/m = GM/r^2$	Radial field. gr^2 is constant, which you can use to test data from a graph. Units: N kg^{-1} (= m s^{-2}).
$V_{\text{grav}} = -GM/r$	Zero at infinity, negative everywhere else. Read (r , V) from a graph to find $M = -rV/G$.
$\Delta E_{\text{grav}} = m\Delta V = GMm(1/r_1 - 1/r_2)$	Never use $mg\Delta h$ over large distances; the schemes award nothing for it because g is not constant.
$GMm/r^2 = m\omega^2r = mv^2/r$	with $\omega = 2\pi/T$ or $v = 2\pi r/T$, giving $T^2 = 4\pi^2r^3/GM$. A memorised Kepler result gets full credit, but the derivation is often asked (3 marks).

Escape speed $v = \sqrt{2GM/R}$	From $\frac{1}{2}mv^2 = GMm/R$. Set $v = c$ for a black-hole radius: $R = 2GM/c^2$ (Oct 2025).
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ROUTINE: “DOES IT MAKE 16 ORBITS A DAY?” / “IS IT GEOSTATIONARY?” (4-5 MARKS)

1. Equate GMm/r^2 to $m\omega^2r$ (or mv^2/r). Remember $r = R_E + h$.
2. $\omega = \sqrt{GM/r^3}$, then $T = 2\pi/\omega$.
3. Orbits per day = $86\,400 \text{ s} \div T$ (a whole number if asked for “complete orbits”).
Compare with the claim and conclude.

Geostationary: $T = 24 \text{ h}$, so $r = (GMT^2/4\pi^2)^{1/3} = 4.23 \times 10^7 \text{ m}$ and height = $3.6 \times 10^7 \text{ m}$. It must be equatorial and move in the same direction as Earth’s rotation.

EXAMPLE (JUN 2025). 6500 kg satellite raised from the surface to geostationary orbit: $\Delta V = GM(1/6.37 \times 10^6 - 1/4.22 \times 10^7) = 5.32 \times 10^7 \text{ J kg}^{-1}$, so $\Delta E = 3.5 \times 10^{11} \text{ J}$. The scheme withheld the mark if the satellite’s radius was wrongly added.

Explanations that recur

Gravitational vs electric fields	Similar: both follow an inverse square law (radial fields), both have infinite range. Different: gravity is only attractive and acts on mass; electric force can be attractive or repulsive and acts on charge.
Equipotentials get further apart	$V \propto -1/r$. Equal steps of ΔV (equal work) need larger Δr further out because the field (force) is weaker there.
Lower orbit → shorter period	$T^2 = 4\pi^2 r^3/GM$, with G and M constant, so $T \propto r^{3/2}$. Or: the force is larger at small r , so ω is greater.
“Astronauts are weightless” is wrong	Weight = mg , and g is still about 8.6 N kg^{-1} at ISS height, so a gravitational force acts. They are in free fall: gravity provides the centripetal force, so there is no contact force and they <i>feel</i> weightless.
Satellite heats on re-entry	Large drag force, so work is done on the satellite (or the air in front is compressed and heats it), and its temperature rises.
Low polar orbit vs geostationary	Advantage: covers the whole surface and gives better resolution because it is closer. Disadvantage: it must be tracked and cannot view one place continuously.
Sun loses mass → planet’s period?	Smaller gravitational force, and gravity provides the centripetal force. With $F = m\omega^2r$, ω decreases so T increases (or the orbit radius increases).
Measuring a mass in orbit	Use an oscillating chair on springs: $m = kT^2/4\pi^2$. Subtract the chair’s mass (Jan 2024).

6. Space: stars, distances and cosmology

$\lambda_{\text{max}}T = 2.898 \times 10^{-3} \text{ m K}$	Wien. Hotter means a shorter peak wavelength. From frequency, first use $\lambda = c/f$.
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$L = \sigma AT^4$, $A = 4\pi r^2$	Stefan-Boltzmann. Same radius: $L \propto T^4$. Same T : $L \propto r^2$.
$I = L/4\pi d^2$	Intensity (W m^{-2}) is what we measure ; luminosity (W) is the total power emitted.
$z = \Delta\lambda/\lambda \approx \Delta f/f \approx v/c$	Laboratory (rest) wavelength in the denominator. Positive $\Delta\lambda$ means redshift and receding; negative means approaching (Andromeda).
$v = H_0 d$, age $\approx 1/H_0$	Convert $\text{km s}^{-1} \text{ Mpc}^{-1}$ to s^{-1} if needed; $1/H_0$ in s $\div 3.15 \times 10^7$ gives years ($\approx 1.4 \times 10^{10}$ yr). $1/H_0$ is the age , not the size.

ROUTINE: THE STAR CHAIN (5 MARKS, ALMOST EVERY PAPER)

Peak λ from a graph or $c/f \rightarrow$ **Wien** $\rightarrow T \rightarrow$ **Stefan** with $A = 4\pi r^2 \rightarrow L$ (or r) $\rightarrow$ **inverse square** $\rightarrow I$ or $d \rightarrow$ compare with the claim. Each arrow is one mark. Square and fourth-power errors are the usual slip, so write $r = \sqrt{(L/4\pi\sigma T^4)}$ explicitly.

EXAMPLE (JUN 2023). Red dwarf: $\lambda_{\text{max}} = 850 \text{ nm} \rightarrow T = 3410 \text{ K}$; $L = 4\pi(1.18 \times 10^8)^2 \times 5.67 \times 10^{-8} \times 3400^4 = 1.33 \times 10^{24} \text{ W} = 0.35 \% \text{ of } L_{\text{Sun}} < 0.5 \%$, so the statement is correct.

EXAMPLE (OCT 2021, SOLAR ROTATION). The $\text{H}\alpha$ line from the two edges of the Sun differs by 0.0040 nm about 656.2797 nm . One edge velocity: $v = c\Delta\lambda/\lambda = 1830 \text{ m s}^{-1}$; $T = 2\pi r/v = 27.8$ days. If the total spread between the receding and approaching edges is given, **halve it** first (Jan 2025).

H-R diagram

- **Axes:** luminosity (L/L_{Sun} , about 10^{-4} to 10^6 , log scale) against surface temperature in K on a **reversed logarithmic** scale (about 40 000 K on the left down to 2500 K on the right). Label at least 2–3 values; the Sun is at $L = 1$, $T \approx 6000 \text{ K}$.
- **Regions:** main sequence runs diagonally from top left to bottom right (mass increases *up* the main sequence). Red giants are top right (cool, large, luminous); red supergiants are above them. White dwarfs are bottom left (hot, small, dim). Red dwarfs are at the bottom right of the main sequence (low L , low T).
- **White dwarf properties:** small radius, very dense, very high surface temperature, low luminosity, no fusion.
- **Cluster age:** a young cluster shows only a main sequence. Red giants (top right), white dwarfs (bottom left) and a missing upper main sequence show the cluster is old, because massive stars leave the main sequence first.
- **Why massive stars leave first:** higher core temperature means hydrogen is fused at a much greater rate, so less time is spent on the main sequence.
- **Placing a star from data (Oct 2025):** lower T than the Sun $\rightarrow$ to the right; larger $r/L \rightarrow$ higher up.

Distance ladder

Trigonometric parallax (3-4 marks)	Observe the star from opposite ends of the diameter of Earth's orbit (6 months apart). Measure the change in angular position against the background of distant ("fixed") stars. The radius of Earth's orbit must be known; use trigonometry ($\tan \theta = r_{\text{orbit}}/d$), not Pythagoras . An annotated diagram can earn the marks.
Limit of parallax	For distant stars the parallax angle is too small to be measured (the percentage uncertainty is too large). Test it by calculating the angle and comparing with the instrument's smallest measurable angle (Jun 2026).
Standard candle (definition, 1 mark)	"An (astronomical/stellar) object of known luminosity ." Do not write brightness or intensity, and do not add "known distance" (that loses the mark).
Standard-candle method (3-4 marks)	Locate a standard candle (Cepheid variable, type 1a supernova) in the galaxy or cluster. Measure (not "calculate") the intensity received on Earth. Its luminosity is known (Cepheid: from its period). Use $I = L/4\pi d^2$ to find d . Dust between us and the star lowers I , so the calculated distance is too large.
Cepheid data (Jan 2025)	Read the period over at least two cycles $\rightarrow$ luminosity from the period-luminosity graph ($\times L_{\text{Sun}}$) $\rightarrow d = \sqrt{L/4\pi I}$.
Hubble method / measuring H_0	Measure the redshift of the galaxy $\rightarrow$ recession velocity $v = zc \rightarrow d = v/H_0$. To find H_0 : plot v against d (distances from standard candles); the gradient is H_0 and the age is $1/H_0$.

Cosmology

- **Redshift:** "the fractional increase in wavelength received, due to the source moving away from the observer."
- **Absorption lines:** electron energy levels are discrete, so a photon is absorbed only if its energy equals the difference between two levels. Only certain transitions are possible, so only certain wavelengths are missing.
- **Exoplanet detection:** the planet exerts a gravitational force on the star, so the star's velocity relative to Earth changes, giving a periodically varying Doppler shift.
- **Dark matter:** galaxy rotation curves show orbital speeds (or orbital periods much shorter than predicted) that need more mass, and so more gravitational force, than the visible matter provides. Dark matter has mass (exerts gravitational force) but **does not emit electromagnetic radiation**; "light" was not accepted.
- **Fate of the universe** depends on mean **density** (not mass) compared with the critical density: greater $\rightarrow$ closed (eventually contracts); less $\rightarrow$ open (expands forever); equal $\rightarrow$ flat. We cannot be sure because the amount of dark matter, and so the density, is uncertain.

7. Oscillations

SHM DEFINITION: 2 MARKS IN ABOUT HALF OF ALL PAPERS. LEARN WORD FOR WORD.

“There is a (resultant) **force/acceleration proportional to the displacement from the equilibrium position** (1) and it **always acts towards the equilibrium position** / in the opposite direction to the displacement (1).” Write “equilibrium position”; the word on its own is rejected. Writing $a = -\omega^2 x$ counts only if every symbol is defined and the minus sign is explained.

$a = -\omega^2 x$, $\omega = 2\pi f = 2\pi/T$	An a - x graph is a straight line through the origin with negative gradient $= -\omega^2$.
$x = A \cos \omega t$, $v = -A\omega \sin \omega t$	$v_{\max} = \omega A$ at the equilibrium position; $a_{\max} = \omega^2 A$ at the amplitude.
$T = 2\pi\sqrt{m/k}$, $T = 2\pi\sqrt{l/g}$	Find k first from the static extension: $k = mg/\Delta x$. Use the total oscillating mass (car + passengers, chair + astronaut).
$E_{k,\max} = \frac{1}{2}m(\omega A)^2$	Alternatively, $E_{k,\max} = \text{elastic PE stored} = \frac{1}{2}F\Delta x = \frac{1}{2}k\Delta x^2$ (energy conserved).

- **Phase:** v leads x by a quarter cycle (sketch v from the gradient of x - t). a and F are in antiphase with x . Given an a - t graph, x is the **inverted** (minus) version of it.
- **Energy graphs:** E_k - t oscillates at **twice** the frequency of x , is always ≥ 0 , and is a maximum when $x = 0$. E_k - x is a parabola, maximum at $x = 0$ and zero at $\pm A$. The total energy is constant without damping.
- **Pendulums:** halving l for the same amplitude doubles $E_{k,\max}$, because $\omega^2 \propto 1/l$ (Oct 2024). Pendulum on Venus: $T_V = T_E\sqrt{(g_E/g_V)}$.
- **Finding k from a graph:** $f^2 = (k/4\pi^2)(1/m)$, so the gradient of f^2 against $1/m$ is $k/4\pi^2$. Measure 30 oscillations, divide, and repeat.
- **Two springs in parallel:** two identical springs side by side double k (Jun 2023: T gave 41.8 N m^{-1} , so each label of 20.9 N m^{-1} was correct).
- **Load limit:** a spring supports less load when oscillating than when static, because below equilibrium the extension (and force) is greater than at equilibrium.

Free, forced, resonance, damping

Resonance (3 marks)	The system is driven/forced at its natural frequency (driving frequency = natural frequency), so there is a maximum transfer of energy , so the amplitude increases to a maximum . Always relate it to the context (cone, car body, bridge, spring balance).
Not resonance (microwave)	The water molecule's natural frequency is about 10 GHz; the oven's driving frequency is 2.45 GHz, which is not at or near it, so this is not resonance.

Damping (2 marks)	Energy is transferred away from the oscillating system (dissipated to the surroundings, increasing the dampers' thermal energy store) because work is done against resistive forces , so the amplitude decreases over time. Damping occurs at all frequencies. Lighter damping means the amplitude decreases more slowly.
Oscillating in water/oil	Viscous drag forces remove energy, so the motion is damped and the amplitude decreases.
Building/bridge dampers	The natural frequency of the damper is matched to the building's (driving) frequency for maximum energy transfer into the damper. A heavy box (large inertia) is strongly damped. Oil is forced through holes, so work is done and energy is transferred to the surroundings and not returned.
Ductile/plastic materials	They deform plastically and absorb energy without returning it; elastic materials return the energy to the oscillation. Plastic: does not return to its original shape when the force is removed.
"Energy is lost at resonance" (student claim)	Energy is conserved in a closed system. Some energy goes to the mass and some to the surroundings; at resonance the least goes to the surroundings.
Insect wing (multiple frequency)	The wing frequency is a multiple of the muscle frequency, so impulses arrive at the same point in each cycle and energy transfer is still efficient.

EXAMPLE (JAN 2023, CAR ON BUMPS). $k = 65 \times 9.81/0.025 = 2.55 \times 10^4 \text{ N m}^{-1}$; $T = 2\pi\sqrt{(1365/2.55 \times 10^4)} = 1.45 \text{ s}$; resonance when the time between bumps = T , so $u = 25 \text{ m}/1.45 \text{ s} = 17 \text{ m s}^{-1}$.

8. Six-mark model answers

Each answer below contains all six indicative points in order, phrased as the schemes phrase them. Learn the chain, not the exact words.

A. GAS PRESSURE RISES WHEN HEATED (OR FALLS WHEN COOLED), KINETIC THEORY. ASKED IN 2020, 2023, 2025.

1. As the temperature increases, the mean kinetic energy of the gas molecules increases,
2. so the mean (rms) speed, and so the momentum, of the molecules increases.
3. The change of momentum of a molecule when it collides with the container wall increases,
4. and molecules collide with the walls more frequently (rate of collision increases; **collisions with the walls**, not with each other).
5. So the rate of change of momentum increases, and the force on the walls increases ($F = \Delta p / \Delta t$).
6. Since $p = F/A$, the pressure exerted by the gas increases (a balloon expands).

B. CONDITIONS FOR FUSION IN A STAR. ASKED IN 2021, 2024 (TWICE), 2026; ALSO AS 4-MARK PARTS.

1. There must be a very high temperature in the core,
2. to give the **nuclei/protons** enough kinetic energy
3. to overcome the electrostatic repulsion between the positively charged nuclei,
4. so that the nuclei get close enough to fuse (close enough for the strong force to act).
5. There must be a very high density (pressure), produced by gravitational forces,
6. to give a collision rate high enough to sustain fusion (and maintain the high temperature).

Write “nuclei” or “protons” throughout. In January 2026 the first use of “atoms/particles/molecules” lost that point. Fusion releases energy because the mass decreases (B.E. per nucleon increases).

C. EVOLUTION OF A SUN-LIKE STAR (WITH H-R POSITIONS). ASKED IN 2021, 2022, 2024, 2025, 2026.

1. The star is on the main sequence (A), fusing hydrogen into helium in its core.
2. When the hydrogen in the core is used up, the rate of fusion decreases or stops,
3. so the core contracts/collapses under gravitational forces,
4. and the core temperature rises until it is high enough for helium fusion to start. (The temperature rise comes *before* helium fusion; the scheme checks the order.)
5. The outer layers expand and cool, and the star becomes a red giant (B, top right): larger surface area, lower surface temperature.
6. When helium runs out and no further fusion takes place, the core collapses to a white dwarf (C, bottom left): small, hot, dim.

Variant (Jan 2021): main-sequence stars move off to the red giant region **above** the main sequence, starting with the most massive (top of the main sequence) first; they end as white dwarfs **below** it.

D. RESONANCE AND DAMPERS (FOOTBRIDGE OR BUILDING). ASKED IN 2025; PARTS APPEAR YEARLY.

1. People walking (or the wind) force/drive the bridge to oscillate with a driving frequency.
2. Resonance occurs when the driving frequency equals the natural frequency of the bridge,
3. so there is a maximum (efficient) transfer of energy to the bridge,
4. and the amplitude of oscillation increases to a maximum (which can be dangerous).
5. Dampers are fitted: energy is transferred from the bridge to the dampers,
6. which dissipate it to the surroundings (work is done against resistive forces), limiting the amplitude.

E. INTERNAL ENERGY OF COOLING, SOLIDIFYING WAX. ASKED IN 2023.

1. The internal energy of the wax decreases throughout the cooling.
2. Internal energy is the sum of the kinetic and potential energies of the molecules.
3. While the temperature falls, the molecular kinetic energy decreases.
4. Between X and Y (flat section) the liquid is solidifying and the molecular potential energy decreases,
5. but the temperature is constant, so there is no change in molecular kinetic energy.
6. At Y the wax has completely solidified (after which the temperature and KE fall again).

F. DETECTING A PARTICLES BY THE UV LIGHT THEY PRODUCE IN AIR. ASKED IN 2021; RARE, BUT USE THE ENERGY-LEVEL CHAIN.

Energy from the α particles is transferred to air molecules; an electron is excited to a higher energy level; when it returns to a lower level a UV photon is emitted; α is strongly ionising so has a short range in air; UV is weakly ionising with a long range; so the UV can be detected far from the source, which is safer.

9. Short-answer phrase bank (1-4 marks)

Main-sequence star	Fusing hydrogen into helium in its core.
Fission	A large (massive) nucleus splits into smaller fragments (+ neutrons). Binding energy per nucleon increases, so mass decreases and energy is released.
Data logger: when useful	Data over a very short (or very long) time interval, or several data sets recorded simultaneously.
Upthrust (balloon, airship)	Upthrust = weight of air displaced. If upthrust > weight there is a resultant upward force; if equal, the forces are balanced and it floats.
Ice in a cool box	Latent heat is needed to melt the ice, so energy is transferred from the food/surroundings to the ice.
Parachute on Mars	The drag force reduces landing velocity, so the rate of change of momentum on impact and the impact force are smaller. A larger parachute is needed because Mars' atmosphere is less dense.
λ_{max} in IR, but star looks red	$\lambda_{\text{max}} > 750 \text{ nm}$ (infrared), but a range of wavelengths is emitted around λ_{max} ; the most intense visible part is red.
Filament lamp L from σAT^4 differs	The filament is not a perfect black body, or the glass absorbs some radiation.
Hot planet loses atmosphere	Higher T means higher mean KE of the molecules, so more molecules exceed the escape velocity.
Intensity from a source	Obeys the inverse square law, so intensity at the outer planets is too small for solar cells and a radioisotope power source is used instead.

10. Multiple-choice traps seen in these papers

- Fate of the universe is decided by **density**, not mass.
- $1/H_0$ gives the age, not the size, of the universe.
- Damping and energy dissipation happen at **all** frequencies; only the maximum energy transfer is special to resonance.
- A forced oscillator vibrates at the **driving** frequency, not its natural frequency.
- E_k is a maximum (never zero) at equilibrium; doubling the amplitude gives $4\times$ the energy.
- "Random" $\neq$ "spontaneous". Which nucleus decays next cannot be predicted.
- B.E. per nucleon increases in **both** fission and fusion.
- α decay energy = $[m_X - m_Y - m_\alpha]c^2$.
- Background is already in the count, so it must still be subtracted. Correct for it before applying the inverse square law.
- Absorption is exponential: two half-thicknesses give $1/4$, not 0.
- $\Delta E_{\text{grav}} = m\Delta V$ is negative moving inwards. As the Moon recedes, the force decreases and the GPE increases.
- Same $p, V, T \rightarrow$ same N and same mean KE; the rms speed differs with molecular mass. $T \times 2 \rightarrow c_{\text{rms}} \times \sqrt{2}$.

- $g \propto M/r^2$; for the same volume, $g \propto$ density.
- H-R: the temperature axis is reversed. Red dwarf = low L , low T . Smaller $\lambda_{\max}$ = hotter.
- Star X hotter and with greater intensity: it may simply be closer. Intensity depends on L **and** d .
- Elastic: returns to original shape. Ductile: deforms plastically and absorbs energy.
- Both fields follow $1/r^2$; only the electric force can repel.
- Specific latent heat experiment: heater on for longer than recorded $\rightarrow L$ too small; energy lost $\rightarrow L$ too large.
- The gradient of v - t gives a (and so F); the gradient of x - t gives v .

11. Final checklist before the exam

Can you, without notes...	Can you, without notes...
Convert $^{\circ}\text{C} \rightarrow \text{K}$, $\text{u} \rightarrow \text{kg}$, $\text{J} \leftrightarrow \text{MeV}$, days/years $\rightarrow$ s?	Write the SHM definition and the resonance and damping answers word for word?
Do a mass-defect calculation to 3 s.f. with no intermediate rounding?	Draw x , v , a , E_k against t with the correct phase and frequency?
Solve $A = A_0 e^{-\lambda t}$ for t using $\ln$?	Derive $T^2 = 4\pi^2 r^3 / GM$ and find orbits per day (remembering $R + h$)?
Write a balanced β^- equation with the antineutrino?	Find ΔE with $GMm(1/r_1 - 1/r_2)$, never mgh ?
Chain Wien $\rightarrow$ Stefan $\rightarrow$ inverse square in one go?	Draw and label an H-R diagram (reverse log T , Sun at 1, 6000 K)?
Describe the parallax and standard-candle methods (3-4 marks each)?	Write the four main 6-markers (sections 8A-D) from memory in under 6 minutes?

Data normally given on the formulae sheet: $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$; $k = 1.38 \times 10^{-23} \text{ J K}^{-1}$; $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$; Wien constant $2.898 \times 10^{-3} \text{ m K}$; $c = 3.00 \times 10^8 \text{ m s}^{-1}$; $e = 1.60 \times 10^{-19} \text{ C}$; $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$; $m_e = 9.11 \times 10^{-31} \text{ kg}$; $g = 9.81 \text{ N kg}^{-1}$.